Amazon – Global E-Commerce Intelligence Suite – Operations & Fulfilment Monitoring

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The Operations & Fulfilment Monitoring Dashboard provides an executive-level analytical view of operational efficiency and logistics performance within a global e-commerce business model inspired by Amazon. Built using a synthetic dataset of 100,000 transaction-level records, this dashboard focuses on order throughput, delivery performance, regional efficiency, fulfilment cost optimization, and operational bottlenecks.

Designed in Power BI using a structured star schema data model, the dashboard enables executive leadership, supply chain managers, and operations teams to monitor fulfilment KPIs through dynamic filtering and structured visual storytelling.

Through KPI cards, slicers, trend analysis, geographic mapping, and executive narratives, the dashboard delivers a consolidated and actionable perspective on operational and fulfilment performance.

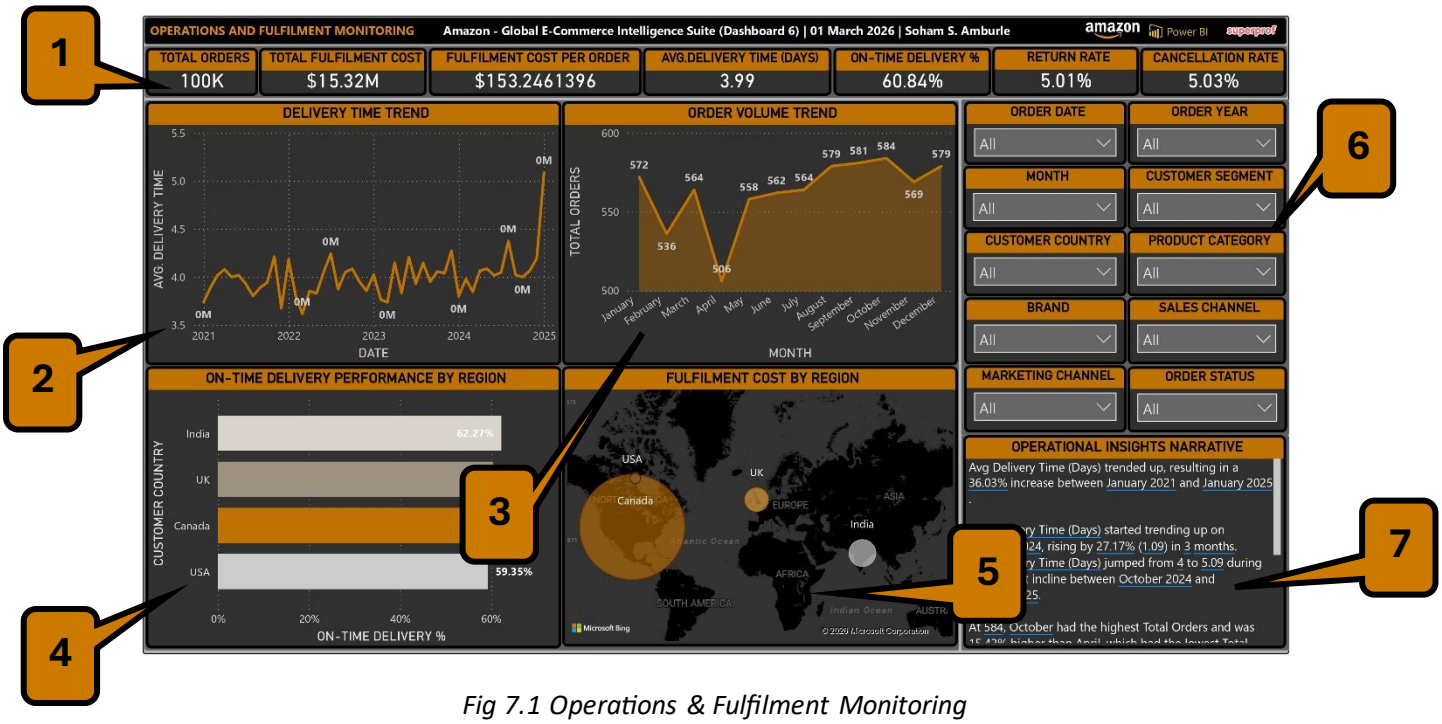

Fig 7.1 Operations & Fulfilment Monitoring

Executive Storyline

The dashboard follows a structured operations analytics narrative:

- Monitor overall order volume and throughput
- Evaluate average delivery times and on-time delivery performance
- Analyze regional fulfilment cost distribution
- Identify return and cancellation trends
- Benchmark operational efficiency metrics
- Provide executive-ready insights through a narrative summary

Data Disclaimer: This project uses a fully synthetic dataset of 100,000 records for demonstration and testing only; no real-world data was used.

This layered structure allows stakeholders to move from high-level KPIs to regional and operational performance insights.

Visuals and Description

1. KPI Cards (Executive Operations Snapshot)

Seven KPI cards provide an immediate summary of operational performance:

1. Total Orders: Total transactions within the selected filter context
2. Total Fulfilment Cost: Overall cost incurred for fulfilment
3. Fulfilment Cost per Order: Average cost per order
4. Average Delivery Time (Days): Average number of days taken to deliver orders
5. On-Time Delivery %: Percentage of orders delivered within target time
6. Return Rate %: Share of orders returned
7. Cancellation Rate %: Share of orders cancelled

These KPIs establish operational scale, cost efficiency, and delivery performance at an executive level.

2. Delivery Time Trend (Line Chart)

X-Axis: Month Name

Y-Axis: Average Delivery Time (Days)

Highlights operational efficiency over time, identifies peak delivery delays, and supports decision-making for process improvements.

3. Order Volume Trend (Area Chart)

X-Axis: Month Name

Y-Axis: Total Orders

Illustrates seasonal trends, peak order periods, and throughput variations.

4. On-Time Delivery Performance by Region (Clustered Bar Chart)

Axis: Customer Country

Value: On-Time Delivery %

Optional Legend: Customer State

Visualizes regional performance differences, enabling targeted operational interventions.

5. Fulfilment Cost by Region (Map Visualization)

Location: Customer Country

Size / Color Saturation: Total Fulfilment Cost

Tooltip: Country/State + Fulfilment Cost

Highlights high-cost regions to optimize logistics strategy and resource allocation.

6. Slicers (Dynamic Filtering Controls)

The dashboard includes ten interactive slicers:

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1. Order Date
2. Year
3. Month Name
4. Customer Segment (Prime / Non-Prime)
5. Customer Country
6. Product Category
7. Brand
8. Sales Channel (Web / Mobile App)
9. Marketing Channel (Google / Meta / Email / Organic)
10. Order Status

All visuals dynamically respond to these filters, enabling multi-dimensional operational performance analysis.

7. Narrative — Operational Insights (Text / Narrative Visual)

Summarizes trends in orders, delivery time, fulfilment cost, returns, and cancellations

Provides executives with actionable insights in plain language

Business Value

This dashboard enables stakeholders to:

- Monitor operational and fulfilment KPIs in a consolidated executive view
- Compare regional efficiency and cost distribution
- Identify bottlenecks in delivery and fulfilment processes
- Track return and cancellation trends
- Support data-driven supply chain and operations decisions

By integrating KPI cards, trend analytics, geographic mapping, slicers, and executive narratives, the dashboard delivers a structured and enterprise-ready operational intelligence solution.

Tools & Data

- Visualization Tool: Power BI
- Dataset: Fully synthetic global e-commerce dataset
- Record Count: 100,000 transaction-level records
- Time Period: 2021–2024
- Model Design: Star Schema (Fact_Orders with Dimension Tables for Date, Customer, Product, and Channel)
- Purpose: Operations and fulfilment analytics demonstration and enterprise BI portfolio development

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